Identifying Patient Heterogeneity in Breast Cancer Through Unsupervised Clustering of METABRIC Gene Expression Data

Abstract

This project explored breast cancer heterogeneity using gene-expression and clinical data from 1,980 samples in the METABRIC dataset. Principal component analysis (PCA) and K-means clustering were applied to 2,000 highly variable genes to identify patient groups with distinct molecular and clinical characteristics.

Two clusters were identified, comprising 1,414 and 566 samples. Cluster 0 was predominantly ER-positive and associated with luminal subtypes, while Cluster 1 contained higher proportions of ER-negative, high-grade, basal-like, and claudin-low tumors. Differences in gene-expression patterns further distinguished the groups. Although cluster assignments were highly stable, separation between the clusters was modest.

Overall, the findings demonstrate how unsupervised clustering can identify established patterns of breast cancer heterogeneity and provide a foundation for further computational analysis.

Table of Contents

Abstract

1. Introduction……..4
2. Data and Methodology…….5
3. Results……10
4. Discussion……..16
5. Conclusion…….19

References……..20

1. Introduction

Breast cancer is a heterogeneous disease characterized by substantial variation in its molecular features, pathological characteristics, and clinical behavior. Tumors that appear similar based on conventional clinical assessments may exhibit distinct biological properties, contributing to differences in disease progression, prognosis, and treatment response. Understanding this heterogeneity is therefore important for improving the molecular characterization of breast cancer and supporting more individualized approaches to disease management.

Gene-expression profiling has contributed significantly to the identification and characterization of breast cancer molecular subtypes. Foundational studies identified distinct expression patterns associated with luminal, HER2-enriched, basal-like, and normal-like tumor characteristics (Perou et al., 2000; Sørlie et al., 2001). Subsequent research further characterized intrinsic subtypes, including luminal A and luminal B, and demonstrated their prognostic and predictive relevance (Parker et al., 2009). These findings established that molecular differences between tumors can provide information beyond conventional pathological characteristics. However, breast cancer remains biologically diverse, and variation in gene-expression patterns may exist even among tumors assigned to the same molecular subtype. Exploring this variation can provide further insight into how individual tumors differ and how those differences relate to established clinical characteristics.

The Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) study demonstrated the molecular diversity of breast cancer and identified distinct subgroups through integrated genomic and transcriptomic analysis (Curtis et al., 2012). Building on this foundation, the present study investigates whether unsupervised analysis of gene-expression profiles can identify patient groups with distinct clinical and molecular characteristics. By examining differences between these groups and genes exhibiting prominent expression differences, the study aims to explore biologically meaningful patterns of patient heterogeneity.

2. Data and Methodology

2.1 Data Source and Study Population

This study utilized publicly available breast cancer molecular and clinical data from the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC) study, accessed through cBioPortal for Cancer Genomics (Curtis et al., 2012; Gao et al., 2013). METABRIC is a large-scale breast cancer research cohort containing molecular profiles and associated clinical information, providing a resource for investigating tumor heterogeneity.

Three datasets were obtained from cBioPortal: gene-expression measurements generated using the Illumina microarray platform, sample-level clinical information, and patient-level clinical information. The clinical datasets included characteristics such as estrogen receptor (ER) status, human epidermal growth factor receptor 2 (HER2) status, tumor grade, molecular subtype, and age at diagnosis.

The downloaded clinical data contained 2,509 patient and sample records, while gene-expression measurements were available for 1,980 tumor samples. Clinical information was linked to gene-expression profiles using the corresponding patient and sample identifiers. The final analytical cohort therefore consisted of 1,980 breast cancer samples with available gene-expression profiles and matched clinical records.

The analysis used the processed gene-expression measurements provided by cBioPortal rather than raw microarray intensity files. All subsequent preprocessing, dimensionality reduction, clustering, and statistical analyses were performed using Python.

2.2 Data Preprocessing

The downloaded METABRIC datasets were processed to create a consistent gene-expression matrix and corresponding clinical dataset for subsequent analysis. Patient-level and sample-level clinical records were combined using their shared patient identifiers (PATIENT_ID). Gene-expression profiles were then matched with the clinical records using sample identifiers (SAMPLE_ID), restricting the analytical dataset to samples with available expression measurements and corresponding clinical information.

The gene-expression dataset was examined for missing values and duplicate gene symbols. A total of 16 missing expression measurements were identified and imputed using the mean expression value of the corresponding gene across available samples. Where multiple expression measurements were associated with the same gene symbol, their values were averaged to obtain a single expression profile per gene. This reduced redundancy while retaining the available gene-level information.

Following preprocessing, the expression matrix was organized with samples as rows and genes as columns, allowing each sample to be represented by its gene-expression profile. The corresponding clinical information was retained separately for subsequent comparisons between the identified patient clusters. No additional normalization of raw microarray intensities was performed, as the analysis used processed expression measurements obtained from cBioPortal.

2.3 Gene Selection

To reduce dimensionality and focus the analysis on genes exhibiting substantial variation across breast cancer samples, variance-based gene selection was performed on the preprocessed expression matrix. The variance of each gene's expression values was calculated across all available samples, and the 2,000 genes with the highest variance were retained for subsequent analysis.

This approach prioritized genes showing greater differences in expression across samples while excluding genes with relatively uniform expression profiles. Selecting a subset of highly variable genes also reduced the computational complexity of subsequent dimensionality reduction and clustering. The resulting expression matrix contained 1,980 breast cancer samples and 2,000 selected genes.

Gene selection was performed without using clinical characteristics or predefined molecular subtype labels, allowing subsequent clustering to be based on variation in gene-expression profiles rather than existing patient classifications.

2.4 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was applied to the expression matrix containing the 2,000 selected genes to reduce dimensionality while retaining major patterns of variation across breast cancer samples. PCA transforms the original gene-expression variables into a set of orthogonal principal components, ordered according to the amount of variance they explain.

The explained variance of individual components and their cumulative contribution were examined to assess the information retained at different levels of dimensionality reduction. The first 20 principal components were selected for subsequent clustering as an exploratory balance between reducing dimensionality and preserving variation in the expression profiles. PCA was performed directly on the processed gene-expression measurements without additional feature standardization. The first two principal components were used to visualize sample distributions in two dimensions.

2.5 Unsupervised Clustering

K-means clustering was applied to the first 20 principal components to identify groups of breast cancer samples with similar gene-expression patterns. K-means is an unsupervised learning algorithm that partitions observations into a predefined number of clusters by minimizing the within-cluster sum of squared distances to their respective cluster centroids.

To determine an appropriate number of clusters, solutions ranging from two to six clusters ($k = 2–6$) were evaluated using the silhouette score, which measures how closely samples resemble others within their assigned cluster relative to samples in neighboring clusters. Higher silhouette scores indicate better-defined separation between groups.

The solution with the highest silhouette score was selected for subsequent analysis. The final K-means model used a fixed random seed of 67 and 10 initializations to support reproducibility. Cluster assignments were retained for subsequent comparisons with clinical characteristics and gene-expression patterns.

2.6 Clinical and Gene-Expression Analysis

Following clustering, sample assignments were linked to the corresponding clinical records to examine differences in patient and tumor characteristics between the identified groups. The analysis focused on estrogen receptor (ER) status, human epidermal growth factor receptor 2 (HER2) status, tumor grade, molecular subtype, and age at diagnosis.

For categorical variables, including ER status, HER2 status, tumor grade, and molecular subtype, contingency tables were constructed to examine the distribution of each characteristic across clusters. Pearson's chi-square tests were used to assess associations between cluster membership and these

categorical characteristics. Age at diagnosis was summarized using descriptive statistics and compared between clusters using Welch's independent-samples t-test, which does not assume equal population variances. Samples with missing values for a particular clinical characteristic were excluded from the corresponding comparison.

To account for multiple statistical comparisons, the resulting p-values from the five clinical tests were adjusted using the Benjamini–Hochberg procedure to control the false discovery rate (Benjamini & Hochberg, 1995). Both unadjusted and adjusted p-values were retained for interpretation.

To characterize gene-expression differences between clusters, the mean expression of each of the 2,000 selected genes was calculated separately for each group. The difference in mean expression between clusters was then computed, and genes were ranked according to the absolute magnitude of these differences. Genes with the largest positive or negative differences were examined to identify expression patterns associated with each cluster and support preliminary biological interpretation. This analysis was descriptive; no gene-level hypothesis tests or differential-expression significance estimates were performed.

2.7 Cluster Stability Assessment

To evaluate the robustness of the identified patient groups, cluster stability was assessed by examining the consistency of cluster assignments under different analytical settings. The analysis focused on two factors: the random initialization of K-means clustering and the number of principal components retained for clustering.

First, K-means clustering was repeated using six alternative random seeds (10, 20, 30, 42, 50, and 100), while maintaining two clusters and 20 principal components. Second, the analysis was repeated using 10, 20, and 30 principal components to examine the sensitivity of cluster assignments to the dimensionality-reduction settings.

Consistency between clustering solutions was evaluated using the Adjusted Rand Index (ARI), which measures agreement between two sets of cluster assignments while accounting for agreement expected by chance. ARI values can range from negative values to 1, with a value of 1 indicating identical partitions and values near 0 indicating agreement comparable to chance.

These analyses were performed to determine whether changes in clustering settings affected patient group assignments. Although consistent assignments indicate that the clustering results are stable, they do not necessarily mean that the identified patient groups are clearly separated or represent distinct biological subtypes.

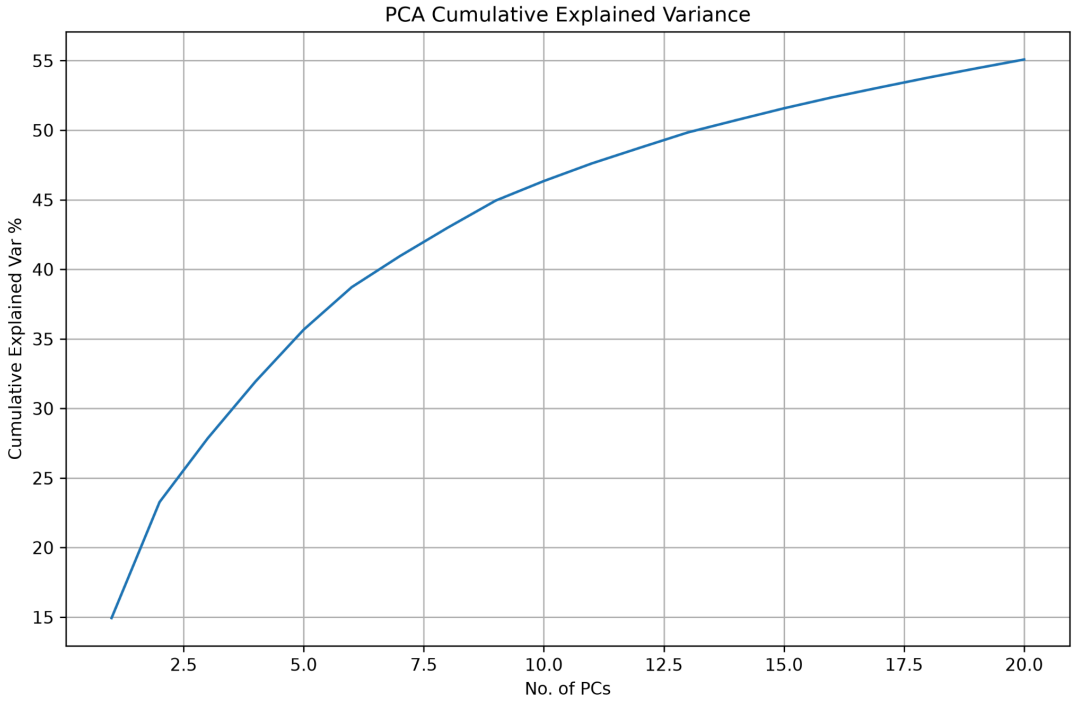
3. Results

3.1 Principal Component Analysis and Variance Explained

Principal Component Analysis (PCA) was performed on the expression profiles of 2,000 highly variable genes across 1,980 breast cancer samples. The first principal component (PC1) explained 14.94% of the total variance, while the second principal component (PC2) explained 8.34%, together accounting for 23.28% of the variation in gene expression.

The cumulative explained variance increased with the number of principal components retained. The first 10 components accounted for approximately 46.35% of the total variance, while the first 20 explained 55.09% (Figure 1). These findings indicate that variation in the gene-expression profiles was distributed across multiple components rather than being concentrated entirely in the first few dimensions.

Figure 1. Cumulative explained variance from PCA of METABRIC gene-expression data.

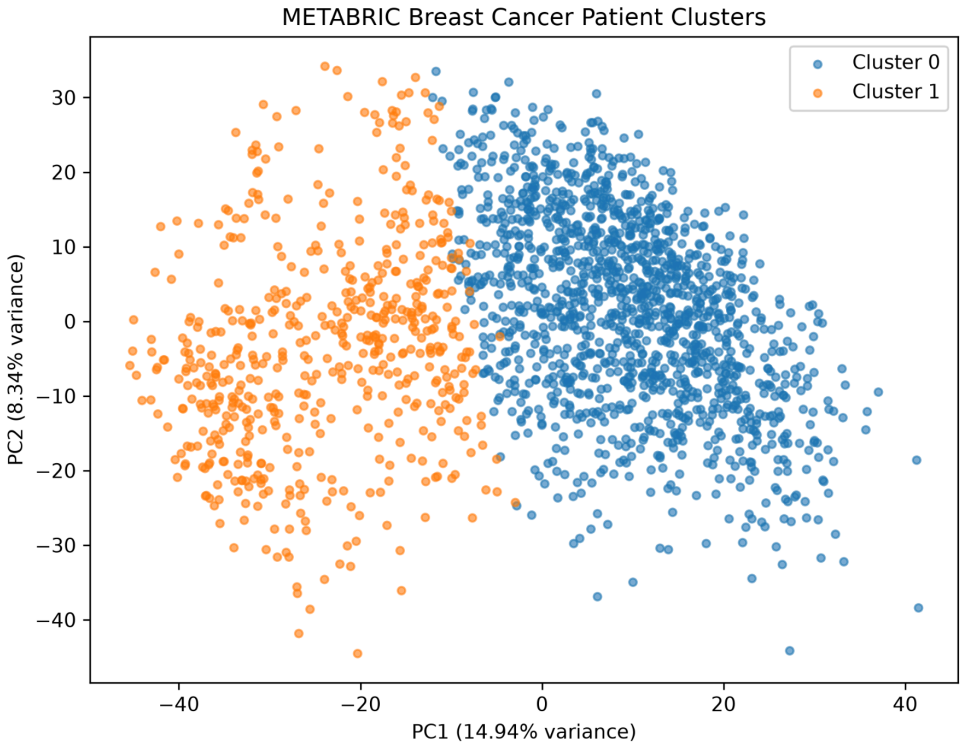


3.2 Identification of Patient Clusters

K-means clustering was evaluated using two to six clusters, with the highest silhouette score obtained for the two-cluster solution (0.219). The silhouette scores for three, four, five, and six clusters were 0.141, 0.145, 0.124, and 0.123, respectively. Based on these results, the two-cluster solution was selected for further analysis.

The final clustering divided the 1,980 breast cancer samples into two groups. Cluster 0 contained 1,414 samples (71.41%), while Cluster 1 contained 566 samples (28.59%). Visualization of the cluster assignments using the first two principal components showed differences in the distribution of the groups, although some overlap was observed (Figure 2). This was consistent with the relatively modest silhouette score, indicating limited separation between the identified clusters.

Figure 2. PCA visualization of the two breast cancer patient clusters.



3.3 Clinical Characteristics of the Clusters

The two patient clusters exhibited substantial differences in clinical and molecular characteristics (Table 1). Cluster 0 was predominantly estrogen receptor (ER)-positive, with 97.81% of samples classified as ER-positive, compared with 21.73% in Cluster 1. HER2-positive tumors were more frequent in Cluster 1 (24.73%) than in Cluster 0 (7.57%).

Tumor grade also differed between the groups. Grade 3 tumors accounted for 82.90% of samples with available grade information in Cluster 1, compared with 37.17% in Cluster 0. Conversely, Grade 1 and Grade 2 tumors were more common in Cluster 0. Differences were also observed in molecular subtype distributions. Luminal A and Luminal B subtypes together represented 80.90% of Cluster 0, whereas basal-like and claudin-low subtypes accounted for 68.73% of Cluster 1.

Patients in Cluster 0 were older at diagnosis, with a mean age of 63.22 years, compared with 55.76 years in Cluster 1. Statistical comparisons identified significant associations between cluster membership and ER status, HER2 status, tumor grade, and molecular subtype, as well as a significant difference in age at diagnosis. All five comparisons remained statistically significant after Benjamini–Hochberg correction (adjusted $p < 0.001$, Table 2).

Table 1. Clinical and molecular characteristics of the patient clusters.

Characteristic	Cluster 0	Cluster 1
Number of samples	1,414	566
ER-positive	97.81%	21.73%
HER2-positive	7.57%	24.73%
Grade 1	11.94%	1.47%
Grade 2	50.89%	15.62%
Grade 3	37.17%	82.90%
Luminal A/B	80.90%	5.48%
Basal/Claudin-low	2.69%	68.73%

Mean age at diagnosis	63.22 years	55.76 years

Note. Tumor-grade percentages exclude samples with missing grade information. Molecular subtype percentages are calculated within each cluster.

Table 2. Statistical comparison of clinical characteristics between patient clusters

Clinical characteristic	Statistical test	Adjusted p-value
ER status	Chi-square	9.315×10^{-280}
HER2 status	Chi-square	3.384×10^{-25}
Tumor grade	Chi-square	1.623×10^{-71}
Molecular subtype	Chi-square	2.720×10^{-274}
Age at diagnosis	Welch's t-test	1.116×10^{-29}

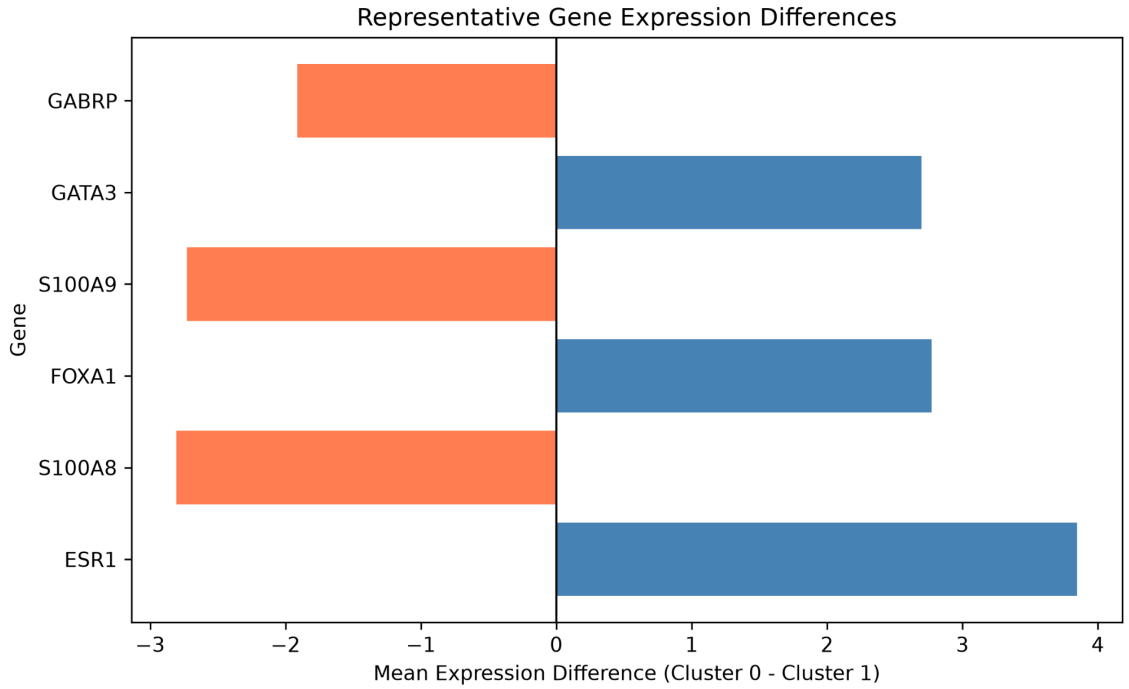
3.4 Gene-Expression Differences Between Clusters

Comparison of mean gene-expression levels across the two clusters revealed differences in several genes associated with breast cancer biology. Among the 2,000 selected genes, ESR1, AGR3, TFF3, TFF1, FOXA1, and GATA3 exhibited higher mean expression in Cluster 0. ESR1 showed the largest positive mean expression difference (3.85), followed by AGR3 (3.79) and TFF3 (3.56).

In contrast, S100A8, S100A9, RARRES1, PROM1, and PSAT1 exhibited higher mean expression in Cluster 1. The largest differences in this direction were observed for S100A8 (−2.81), S100A9 (−2.73), and RARRES1 (−2.41). These differences highlight distinct gene-expression patterns between the identified patient groups (Figure 3).

The reported values represent differences in mean expression between Cluster 0 and Cluster 1, rather than fold changes or statistically significant differential-expression estimates.

Figure 3. Gene-expression differences between patient clusters.



3.5 Cluster Stability Results

Cluster stability was evaluated by repeating K-means clustering with different random seeds and varying the number of principal components used for clustering. The results demonstrated a high degree of consistency in patient assignments across the tested settings.

Using six alternative random seeds (10, 20, 30, 42, 50, and 100), the Adjusted Rand Index (ARI) was 1.000 for every comparison, indicating identical cluster partitions. Similarly, changing the number of principal components to 10 or 30 produced ARI values of 0.998 relative to the 20-component reference solution, demonstrating minimal changes in cluster membership.

A comparison between the original saved cluster assignments and the clustering solution reproduced during validation yielded an ARI of 0.996, with 1,978 of the 1,980 samples (99.90%) assigned consistently. Only two samples differed between the solutions.

Overall, these findings indicate that the identified patient groups were highly stable under the tested analytical settings. However, this stability does not necessarily imply strong separation between the clusters or establish their biological distinctiveness.

Table 3. Cluster stability across analytical settings

Validation setting	ARI
Alternative K-means random seeds (six tested)	1.000
10 principal components	0.998
20 principal components (reference)	1.000
30 principal components	0.998
Original vs. reproduced assignments	0.996

4. Discussion

4.1 Interpretation of Identified Patient Groups

The findings of this study suggest that unsupervised clustering of gene-expression profiles can identify patient groups with distinct molecular and clinical characteristics. Although the clustering algorithm did not use clinical information to define the groups, the resulting clusters showed substantial differences in estrogen receptor status, HER2 status, tumor grade, molecular subtype distribution, and age at diagnosis.

Cluster 0 exhibited characteristics commonly associated with luminal breast cancer, including a high proportion of ER-positive tumors, greater representation of Luminal A and Luminal B subtypes, and higher expression of estrogen-related genes. In contrast, Cluster 1 contained a greater proportion of ER-negative, high-grade, basal-like, and claudin-low tumors, alongside higher expression of genes associated with inflammatory and basal-like expression patterns.

These findings indicate that the clustering captured meaningful variation in molecular and clinical characteristics. However, the relatively modest silhouette score suggests that the groups were not strongly separated, highlighting the complexity and overlap present in breast cancer gene-expression profiles.

4.2 Biological Significance of Gene-Expression Patterns

The gene-expression differences observed between the two clusters provide additional insight into their molecular characteristics. Several genes exhibiting higher expression in Cluster 0, including ESR1, FOXA1, and GATA3, are involved in estrogen receptor signaling and luminal breast cancer biology. ESR1 encodes estrogen receptor alpha, while FOXA1 and GATA3 are transcription factors involved in maintaining luminal epithelial identity and regulating estrogen-responsive gene expression (Martin et al., 2021). Their elevated expression is consistent with the predominance of ER-positive and luminal tumors in Cluster 0.

In contrast, Cluster 1 exhibited higher expression of genes such as S100A8 and S100A9, which have been associated with inflammatory signaling and tumor-related immune processes (Chen et al., 2023). These expression patterns, together with the greater representation of high-grade and basal-like tumors, suggest that Cluster 1 captures molecular characteristics distinct from the estrogen-associated profile of Cluster 0.

These expression patterns help characterize the identified clusters but should not be considered independent evidence of distinct biological mechanisms.

4.3 Relationship to Existing Breast Cancer Research

The findings of this study are consistent with previous research demonstrating that breast cancer exhibits substantial molecular heterogeneity. Early gene-expression studies by Perou et al. (2000) and Sørlie et al. (2001) identified distinct molecular patterns associated with luminal, basal-like, and HER2-enriched breast cancers. The differences observed between the two clusters in the present analysis broadly reflect these established patterns, particularly the distinction between estrogen receptor-associated and basal-like tumor characteristics.

The original METABRIC study further demonstrated the complexity of breast cancer by identifying molecularly distinct subgroups through integrated genomic and transcriptomic analysis (Curtis et al., 2012). Unlike the METABRIC study, which incorporated multiple types of molecular information, the present analysis focused on gene-expression data and used a comparatively simple unsupervised clustering approach. Consequently, the two clusters identified here should not be considered equivalent to the molecular subgroups described in the original METABRIC research.

Nevertheless, the observed associations between gene-expression-based clusters and clinical characteristics demonstrate how unsupervised computational methods can recover established patterns of breast cancer heterogeneity without directly using clinical labels during clustering. The results therefore provide an exploratory characterization of patient differences rather than a new molecular classification system.

4.4 Study Limitations

Several limitations should be considered when interpreting the findings of this study. First, although the two-cluster solution achieved the highest silhouette score among the evaluated configurations, the score remained relatively modest. This indicates that the identified patient groups were not strongly separated and that some overlap existed between their gene-expression profiles.

Second, the analysis relied on a single publicly available breast cancer dataset. Although the cluster assignments remained stable under the tested analytical settings, their reproducibility across independent

patient populations was not evaluated. Furthermore, the analysis used K-means clustering, which may not capture more complex or overlapping structures within gene-expression data.

Third, gene-expression differences were examined descriptively without formal gene-level statistical testing. Because the same expression data were used to construct and characterize the clusters, these comparisons cannot serve as independent validation. Similarly, the molecular subtype labels were derived from gene-expression-related information and therefore do not provide fully independent evidence of biological distinctiveness.

Finally, the study focused on associations between cluster membership and selected clinical characteristics. It did not examine patient survival, treatment response, or disease progression. Consequently, the clinical utility and prognostic relevance of the identified groups remain uncertain.

4.5 Potential Extensions of the Analysis

The present analysis could be extended by applying alternative clustering methods, such as hierarchical clustering, and evaluating whether the identified patient groups remain consistent. Formal differential-expression and pathway enrichment analyses could provide further insight into the molecular differences between the clusters.

Additionally, examining available clinical outcomes, such as survival and disease recurrence, could help assess the clinical relevance of the identified groups. These extensions would provide a more comprehensive understanding of the patterns observed in this exploratory analysis.

5. Conclusion

This project explored breast cancer heterogeneity through unsupervised clustering of gene-expression data from the METABRIC dataset. Using principal component analysis and K-means clustering, 1,980 breast cancer samples were divided into two groups with distinct molecular and clinical characteristics.

The identified clusters exhibited differences in clinical characteristics and gene-expression patterns, broadly reflecting established breast cancer molecular characteristics. Although cluster assignments were highly stable across the tested settings, the modest separation between groups highlights the complexity of patient heterogeneity.

Overall, this exploratory analysis demonstrates how unsupervised computational methods can identify clinically relevant patterns within complex gene-expression data. While the findings largely reflect established breast cancer characteristics, the project provides a foundation for further investigation of patient heterogeneity using more comprehensive analytical approaches.

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